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DEPARTMENT OF

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Subject : Fundamentals of Telecommunication Systems

Experiment Number : 04

Experiment Title : Demonstration of Modem Hardware and Signal Flow

Date of Performance : 09/02/2026

Date of Submission : 09/02/2026

Experiment No. 04: Demonstration of Modem Hardware and Signal Flow

Aim

To identify the hardware interfaces of a Modem and demonstrate the flow between a PC and an ISP Server using Cisco Packet Tracer.

Theory

What is a Modem?

A modem is a hardware networking device that converts data between digital formats used by computers and analog signals used for transmission over telecommunication lines (such as telephone or cable wires). It serves as a bridge between a local network and the internet service provider's infrastructure. While computers process information in binary (zeros and ones), older transmission infrastructure often relies on continuous analog waves; the modem facilitates communication between these two disparate systems.

Explain the working principle of a modem.

The working principle of a modem revolves around two distinct processes: **Modulation** and **Demodulation**.

- **Modulation (Digital to Analog):** When a computer sends data, the modem receives digital signals (binary data). It "modulates" these signals by imposing them onto a carrier wave—changing the wave's amplitude, frequency, or phase to represent the bits. This resulting analog signal can travel effectively across telephone or cable lines.
- **Demodulation (Analog to Digital):** When receiving data from the internet, the incoming signal is analog. The modem "demodulates" this signal, separating the data from the carrier wave and converting it back into the digital binary format that the computer's processor can understand and utilize.

What is IP address?

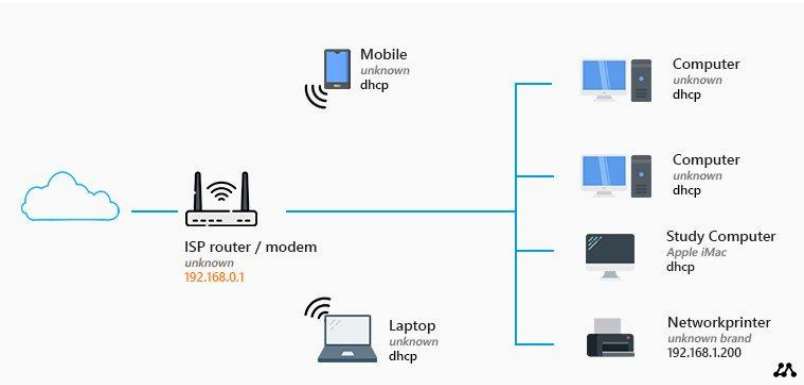
An IP address is a unique numerical label assigned to every device connected to a computer network that uses the Internet Protocol for communication. It serves two principal functions: host or network interface identification and location addressing. Much like a physical home address ensures mail reaches the correct destination, an IP address ensures data packets are routed to the correct device. IP addresses are currently categorized into two versions: **IPv4** (e.g., 192.168.1.1), which uses a 32-bit format, and **IPv6** (e.g., 2001:0db8:85a3...), which uses a 128-bit format to accommodate the growing number of internet-connected devices.

What is ISP?

An Internet Service Provider (ISP) is a company that provides individuals and organizations with access to the internet and related services. ISPs maintain the massive infrastructure of cables, routers, and servers required to route traffic globally. They act as the gateway for users to connect to the global backbone of the internet. ISPs are often categorized by "Tiers":

- **Tier 1** ISPs manage the global backbone.
- **Tier 2** ISPs peer with Tier 1 and other Tier 2 providers.
- **Tier 3** ISPs (local providers) purchase internet access from higher-tier networks to sell directly to end consumers.

Block Diagram



Hardware Components Required

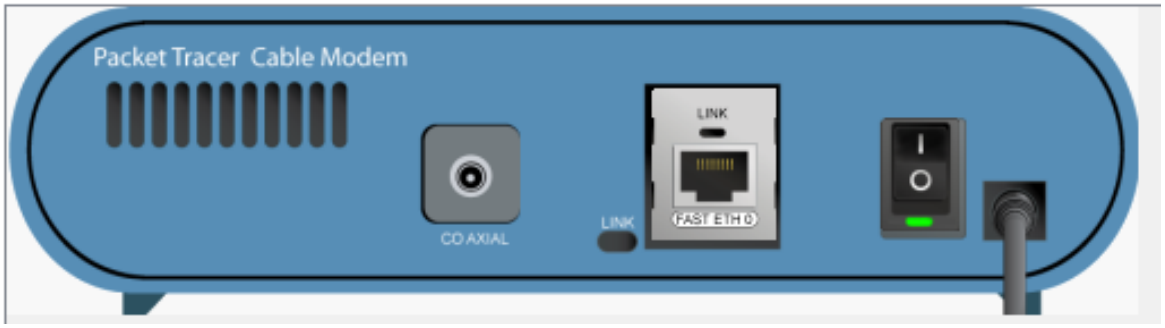
- **PC-PT:** The user's computer (Digital Source).
- **Cable Modem-PT:** The signal converter (Modulator-Demodulator).
- **Cloud-PT (ISP):** The Service Provider infrastructure.
- **Server-PT:** The remote destination (e.g., a Web Server).

Procedure

Step 1: Physical Topology Design

In this stage, we select the hardware from the device box and place them on the logical workspace.

- **Add End Devices:** Go to the bottom-left "End Devices" category. Drag and drop one PC (PC-PT) and one Server (Server-PT) onto the workspace. Rename the PC by your batch and roll number (e.g. S11_01).
- **Add the ISP Infrastructure:** From the "WAN Emulation" category, drag and drop the Cloud-PT between the PC and the Server. Rename this as "ISP_Cloud".
- **Add the Modem:** Go to "Network Devices" -> "WAN Emulation". Drag and drop the Cable Modem-PT between the PC and the Cloud.
- **Hardware Identification:** Click on the Modem and observe the physical rear view. Note the difference between the small RJ-45 Ethernet port and the threaded Coaxial port.



Step 2: Physical Cabling (Media Selection)

Selecting the correct transmission media is crucial for simulating real-world telecommunication links.

- **PC to Modem:** Select the Copper Straight-Through cable (solid black line). Connect the FastEthernet 0 port of the PC to Port 1 (ethernet port) of the Cable Modem.
- **Modem to ISP:** Select the Coaxial cable (solid blue line). Connect Port 0 (co-axial port) of the Cable Modem to the Coaxial 7 port of the ISP_cloud.
- **ISP to Server:** Select the Copper Straight-Through cable. Connect the Ethernet 6 port of the ISP_Cloud to the FastEthernet 0 port of the Server.

Step 3: ISP Cloud Logic Configuration

The Cloud must be told how to "route" the signal from the analog coaxial side to the digital server side.

- Click on the ISP_Cloud and go to the Config tab.
- In the left-hand menu, look under the INTERFACE heading and click on Ethernet6. Ensure "Cable" is selected under the Provider Network.
- In the left-hand menu, look under the CONNECTIONS heading and click on Cable.
- In the drop-down menus, select Coaxial 7 for the first field and Ethernet 6 for the second field.
- Click the Add button. You should now see Coaxial7 <==> Ethernet6 appearing in the mapping table.

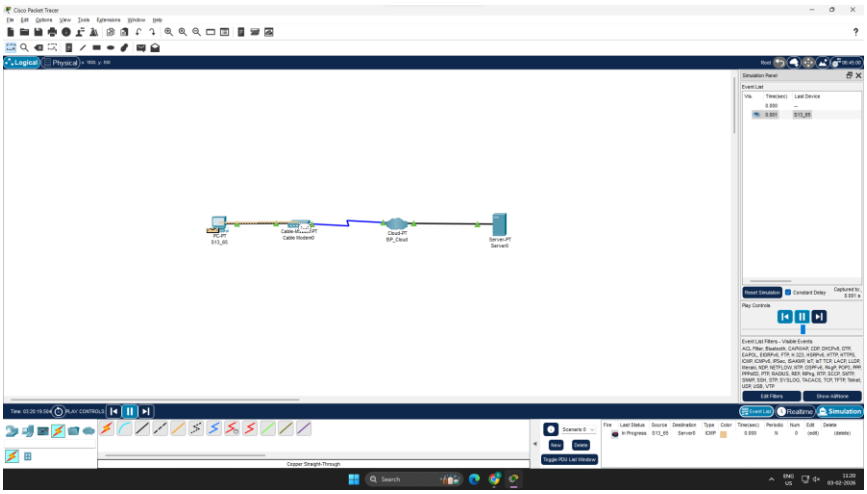
Step 4: IP Addressing (Logical Setup)

Both the source (PC) and destination (Server) must belong to the same logical network to communicate without a router.

- **Configure PC:** Click PC -> Desktop -> IP Configuration. Set the Static IP address to 192.168.1.1 and Subnet Mask to 255.255.255.0.
- **Configure Server:** Click Server -> Desktop -> IP Configuration. Set the IP address to 192.168.1.2 and Subnet Mask to 255.255.255.0.

Step 5: Testing and Packet Observation

- **Real-time Testing:**
  - Open the Command Prompt on the PC. Type ping 192.168.1.2. If you see "Reply from...", the modem is correctly modulating and demodulating the signal.
- **Simulation Mode:** Click the Simulation tab (bottom-right).
  - Click Edit Filters and select only ICMP.
  - Click the Add Simple PDU (Envelope icon), click on the PC, then click on the Server.
  - Click the Play/Forward button to observe the packet travel from the digital Ethernet link to the analog Coaxial link.



Observations

- **Medium Change:** It is observed that the data travels as a digital signal until it reaches the Modem.
- **Signal Conversion:** Upon hitting the Modem, the data is converted to travel over the **Coaxial medium** (simulating analog transmission).
- **Successful Loop:** The Server successfully receives the modulated signal, ISP cloud demaps the return packet back to the specific Coaxial line, allowing the PC to receive a successful reply.

Conclusion

The experiment successfully demonstrates that a Modem is an essential interface for internet connectivity. It proves that digital end-devices cannot communicate directly over analog WAN infrastructure without the modulation/demodulation hardware.